

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Computer Science & Engineering
24 Series

Course No.: 1201
Course Title: Data Structure

Assignment No. : 01

**Assignment Title : Understanding Fundamental Data Structures
and Their Computational Concepts**

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Banking System

Objectives:

- To analyze how fundamental data structures support the internal operations of an ATM-based banking system
- To explain how correct structure selection improves the system's speed, reliability, and resource management

Background:

A banking system maintains sensitive and a huge amount of data, which includes account records, transaction histories (withdrawals and deposits), verification processes and customer requests. For faster verification, secure access, and smooth transaction processing, efficient data structures are necessary, as in peak times, thousands of users access the ATMs simultaneously and performance directly impacts transaction speed and user satisfaction

Functionality	Data Structure	Justifications
Storing account information (Account number, balance, PIN)	Hash Table	Hash Table enables constant-time search without scanning the entire database for account verification using account number as a key
Managing ATM transaction queue	Queue	It ensures "First Come First Served" processing for ATM users to avoid request collisions to maintain fairness and supports real time incoming transactions
Recording user transaction history	Doubly Linked List	Allows efficient insertion of new transaction records at the end while supporting backward and forward traversal

Table: Functionalities and Suitable Data Structures with Justifications

Discussion on Improper Selection:

- Using an array instead of a hash table would require scanning all accounts for every verification which would eventually slow response time
- Using a stack instead of a queue for ATM requests would reverse the service order and disrupt fairness
- Using an array instead of a linked list for transaction logs increases insertion cost and wastes memory

Application & Reflection:

In a banking system, data structures directly affect the speed and stability. Operations such as account verification, withdrawal processing and transaction logging must be executed quickly and consistently. Using a **hash table** for account records makes verification almost instantaneous because the mapping between account number and record eliminates unnecessary searching. The **queue** structure ensures that users at the ATM are processed exactly in the order they arrive, maintaining fairness and preventing request overlap. Transaction histories grow continuously and a **doubly linked list** supports efficient insertion and smooth navigation across records without reallocation

Conclusion:

A banking system runs efficiently when each component uses the appropriate data structure. Correct choices support speed, fairness and system stability. This assignment demonstrated that good system performance comes from choosing structures aligned with the type and pattern of data being processed